

Key words: hippocampal dentate gyrus; chronic stress; glutamate; spatial learning

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P2-69 The protective effects of escitalopram on cognitive behavior in triple-transgenic mice of Alzheimer's disease

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The existence of a high co-morbidity between Alzheimer's disease (AD) and depression has been known for a long time. The antidepressant escitalopram ameliorate the amyloid deposition in AD patients and transgenic animal models. In this study, we firstly tested the effects of escitalopram on behavioral performance in female 3×Tg-AD mice and wild-type mice. Mice were treated with escitalopram (10 mg/kg/day) or the same volume saline by intragastric administration from 9 months of age for 3 months. We found that: (1) escitalopram administration effectively improved short-term and long-term spatial learning and memory of 12-month-old AD mice in Y-maze and Morris water maze tests. (2) Escitalopram ameliorated behavioral disinhibition of AD mice in the elevated plus-maze test. (3) Escitalopram decreased in the duration of immobility of AD mice in the tail suspension test. The present study provides a new evidence for the neuroprotection of antidepressant drugs, suggesting that escitalopram, may be beneficial for the treatment of AD.

Key words: antidepressant; escitalopram; cognitive behavior; 3×Tg-AD mice

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P2-70 Norepinephrine in the hippocampal dentate gyrus is involved in spatial learning and memory alteration induced by chronic restraint stress in aged rats

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